Spatial change-point estimation

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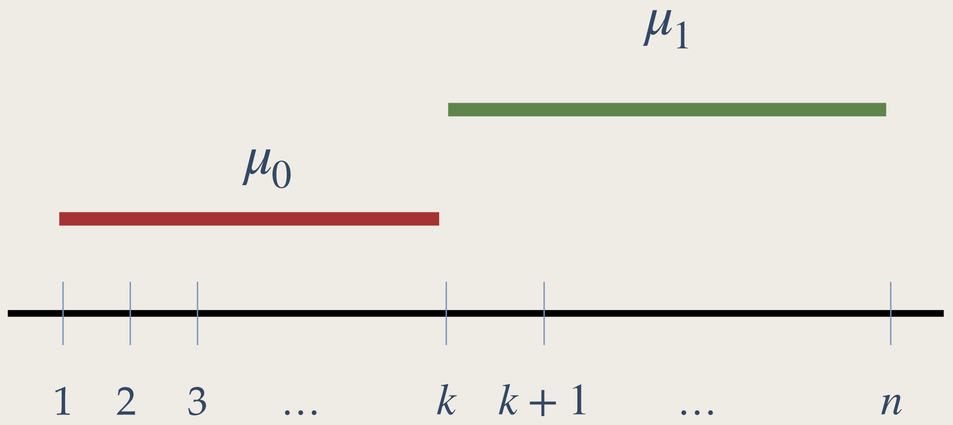
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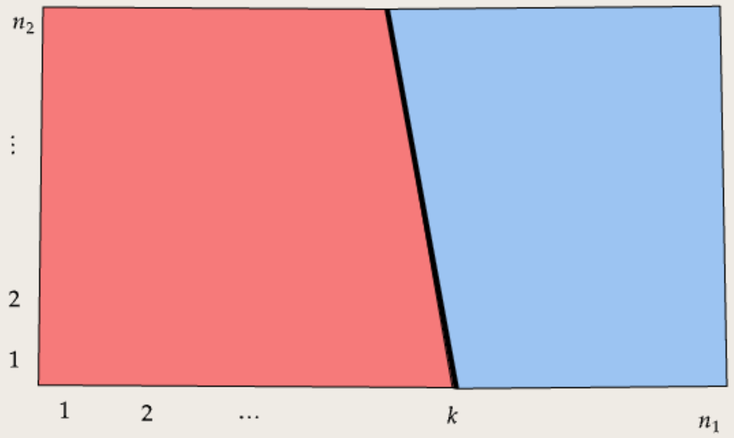
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* *Joint work with* **Sayar Karmakar** (UF), **George Michailidis** (UCLA).

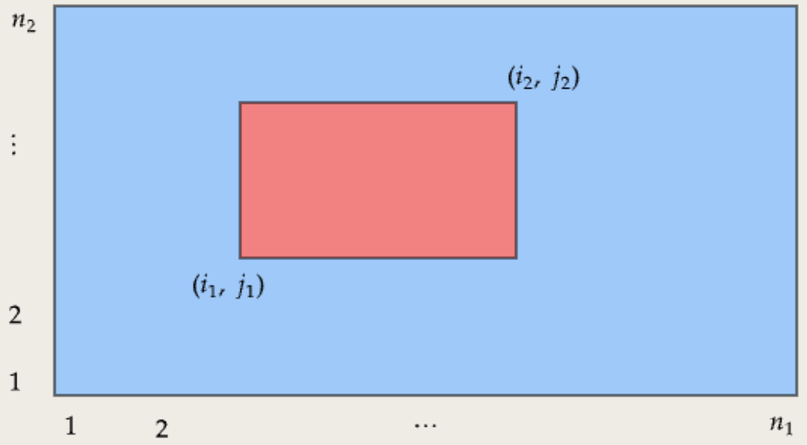
(Mean) Change-point in a time series:



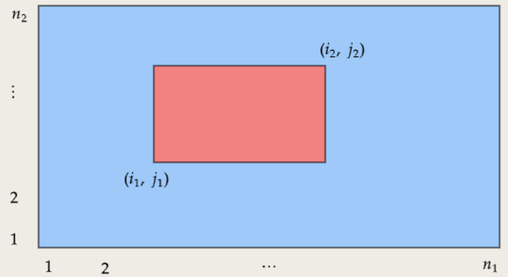
(Mean) Change-(point?) in spatial data:



(A)

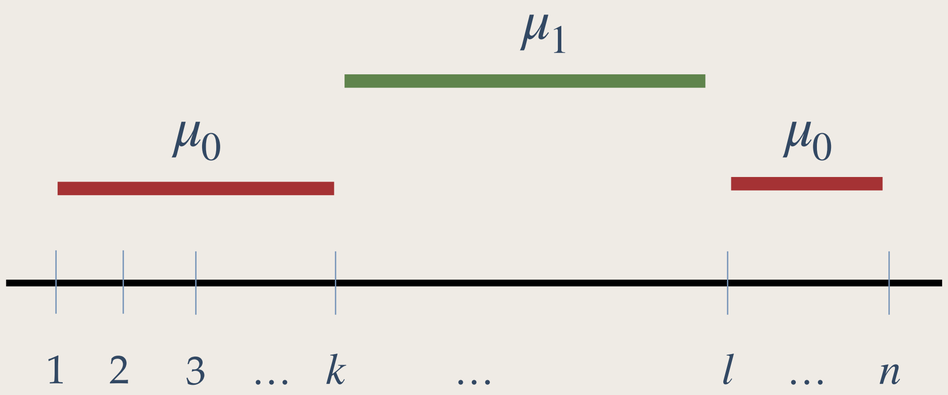


(B)



(B) is more general.

Direct generalization of *epidemic* change-point in time-series*



Easily generalizable to more general patches of change.

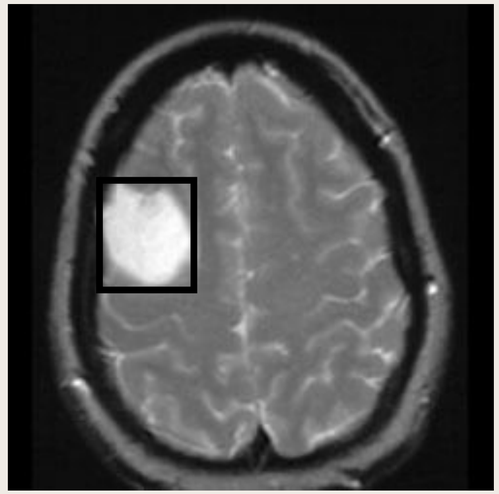


Image from Bucchia and Heuser, JSPI (2015)

* *Huskova (1995, 96), Csorgo-Horvath (1997), Guan (2007), Aston-Kirbarch (2012)*

$$\text{Model: } X_i = \mu_i + \varepsilon_i, \mu_i = \begin{cases} \mu_0, & \text{if } i \in I^c \\ \mu_1, & \text{if } i \in I, I \in \mathscr{R} \end{cases}, \ i \in \Gamma_n \subseteq \mathbb{Z}^d, \ |\Gamma_n| = n$$

- Detecting changes in set-up (B) leads naturally to *scan statistics**: $\max_{R \in \mathscr{R}} T(R)$, where $\mathscr{R}$ is a set of scan windows.
- We focus on when $\mathscr{R}$ is the set of rectangles $I_{[a,b]} = \{x \in \mathbb{Z}^d : a \leq x \leq b\}, a, b \in \mathbb{Z}^d$. Assume the set of sampling locations Γ_n is contained in a rectangle.
  This removes (i) unnecessary technicalities of general patches, (ii) allows localization before applying *computationally expensive algorithms* to detect general patches. More on this later.

* *Glaz-Zhang (2004), Dutta-Sen (2021) along with numerous works by Axel Munk, and Gunther Walther. See Abolhassani-Prates (2021) for a survey.*

- Vast literature; but most/all have one/more of the following limitations.

1. Focus on the testing problem; H_0: There is no change-point patch vs, H_1: $\exists$ at least one change-point patch.

➡ We focus on estimating multiple change-point patches; our results look like:
$\mathbb{P}(\hat{K} = K, \text{ and } |\hat{I}_k \Delta I_k| > M) \approx$ small, where K is the number of true patches $I_1, \ldots, I_K$, and Δ denotes symmetric difference.

2. Use of parametric/ Gaussianity assumption along with independence.

We only assume spatial random field $(\varepsilon_i)_{i \in \mathbb{Z}^d}$ satisfy a Doob-type inequality; for any rectangle $I_{[a,b]}$, it holds that $\mathbb{E}\Big[\Big|\sum_{j \in I_{[a,b]}} \varepsilon_j\Big|^2\Big] \leq C|b-a|$, $|i| = i_1 \ldots i_d$ for $i \in \mathbb{Z}^d$.

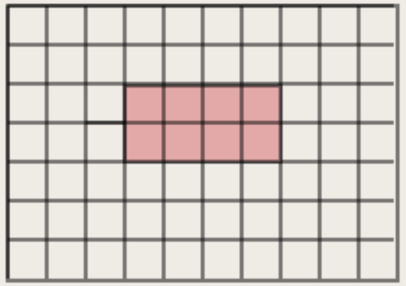
3. The algorithms are usually painfully slow for practical implementation on large spatial datasets.

➡ E.g., if there is only a single rectangle, the usual estimator to localize it is:

$$\hat{I}_{LS} = \arg \min_{I \subset \Gamma_n, n\lambda_2 > |I| > n\lambda_1} \sum_{k \in I} (X_k - \bar{X}_I)^2 + \sum_{k \in I^c} (X_k - \bar{X}_{I^c})^2 = \arg \max_{I \subset \Gamma_n, n\lambda_2 > |I| > n\lambda_1} \sqrt{\frac{|I|(n-|I|)}{n^2}} |\bar{X}_I - \bar{X}_{I^c}| .$$

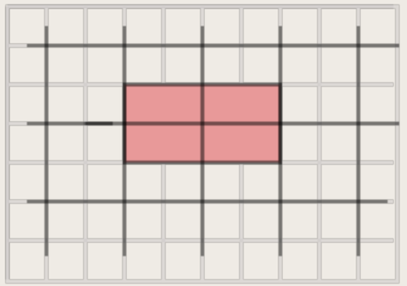
➡ $O(n^2)$ many computations. For context, if $\Gamma_n = \{1, \ldots, m\}^d$, then $n = m^d$.

➡ Can we do better while maintaining consistency?



Original dataset,
red denotes true change – point patch

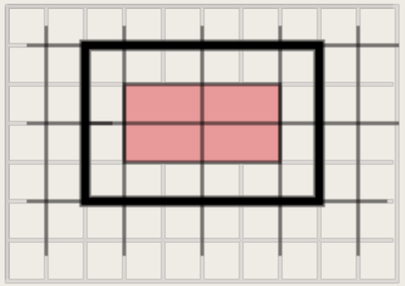
(*Subsample*)



(Suppose $\sqrt{n}$ many subsamples)

(*Least square estimator*)

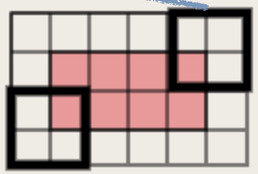
Computations:
$O(\sqrt{n}^2) = O(n)$

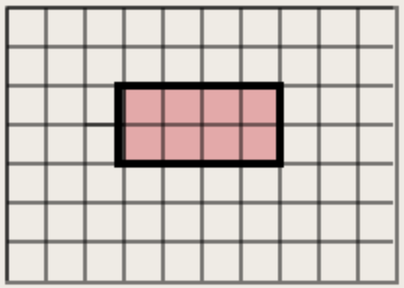


First stage estimator,
based on subsamples.

(*Least square estimator where the search space is restricted to a region around the end – points of the block selected at first stage*)

Local enlargement about naive estimates.





(*Final estimator*)

Computations:
$O(\sqrt{n}^2) = O(n)$

A faster algorithm*!

* *Spatial/dependent generalization of Lu et. Al. (2017).*

- Fact: Statistical accuracy of LS estimates depends on the underlying grid in [0,1] scale.

Here the localization does not change the grid structure, so no loss in accuracy!

Theorem *(Informal, BKM 2025+).*

Let the jump $\delta := |\mu_1 - \mu_0| \geq n^{-\kappa}$; let $c_n = |I|/n$ be the true change-point area in [0,1]-scale. Assume $n^{1-\alpha} c_n^2 \delta^2 \to \infty$. Then, if block size of first stage is n^α and the enlargement size $n^\kappa, \kappa \ll 1 - \alpha$, then under the Doob-type assumption,

$$|\hat{I} \Delta I_0| = O_{\mathbb{P}}(\delta^{-2}) \,.$$

With block sizes n^α, the algorithm takes $O(n^{2\alpha} + n^{2(1-\alpha)})$ computations.

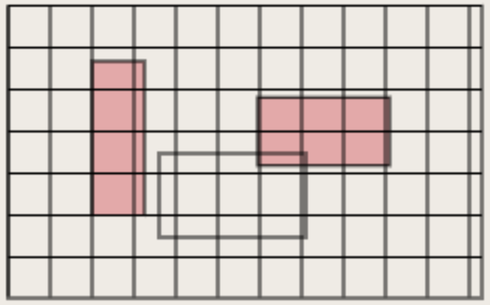
- Key assumptions: $n^{1-\alpha} c_n^2 \delta^2 \to \infty$, $\delta \geq n^{-\kappa}$.

✦ Computationally: $\alpha = 1/2$ is optimal; Statistically, if $c_n \asymp n^{-\gamma}$, we need $\alpha < 1 - 2\gamma$.

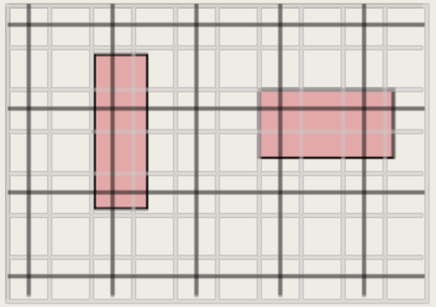
✦ As long as $\gamma < 1/4$, the choice of $\alpha = 1/2$ marries the statistical and computational efficiency.

✦ As γ increases, the size of the change-point decreases; in such cases, α needs to be small, $1 - 2\gamma$ so as to achieve the optimal rate;

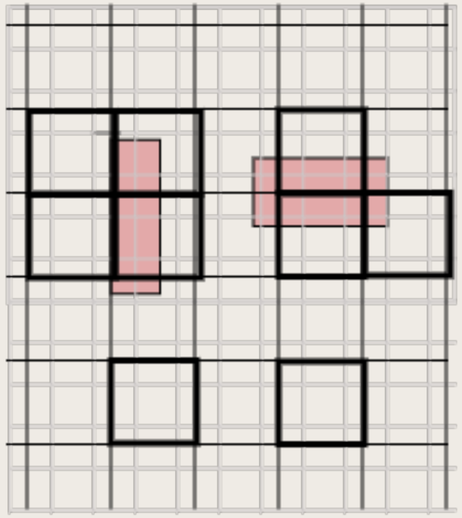
✦ The corresponding increased run-time of $O(n^{4\gamma})$ reflects the increased hardness of the problem under vanishing alternatives.



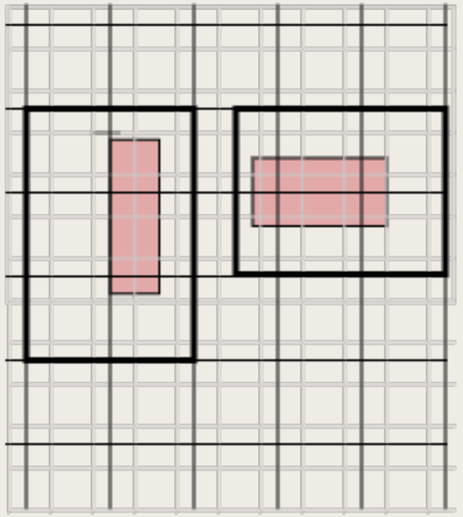
Create blocks



Test for mean at each block

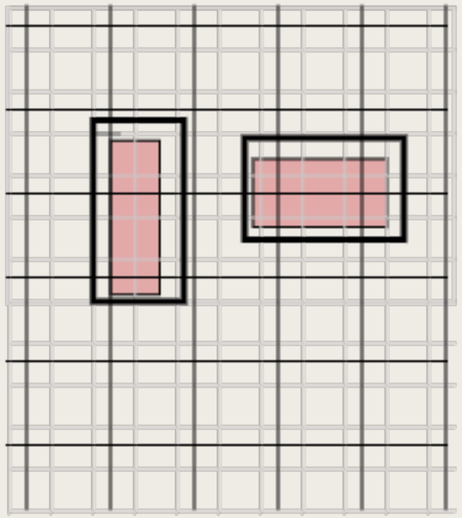


Blocks whose tests were rejected



Consider connected components of selected blocks. Remove smaller isolated components. Enlarge the remaining, irregularly – shaped components into rectangles.

Apply single CP algorithm



Multiple change-patch localization

- To avoid multiple testing issues, we assume a strong invariance principle.

Let $\mathscr{A}$ be the collection of rectangles in $[0,1]^d$. Then, for $A_n := \sum_{i \in \{1,\dots,n\}^d} nA \cap I_{[i-1,i]}$ in an enriched space there exists a standard Brownian Sheet $\mathbb{W}$ on such that

$$\max_{A \in \mathscr{A}} | S_n(A_n) - \sigma n^{d/2} \mathbb{W}(A) | = o_{\mathbb{P}}(n^{d/p})$$

for some $\sigma > 0$ and $p > 2$, where $S_n(A_n) = \sum_{i \in A_n} \varepsilon_i$.

Such results are common in time-series; also proved for liner spatial random field $\varepsilon_i = \sum_s a_s e_{s-i}$, e_j's are independent.

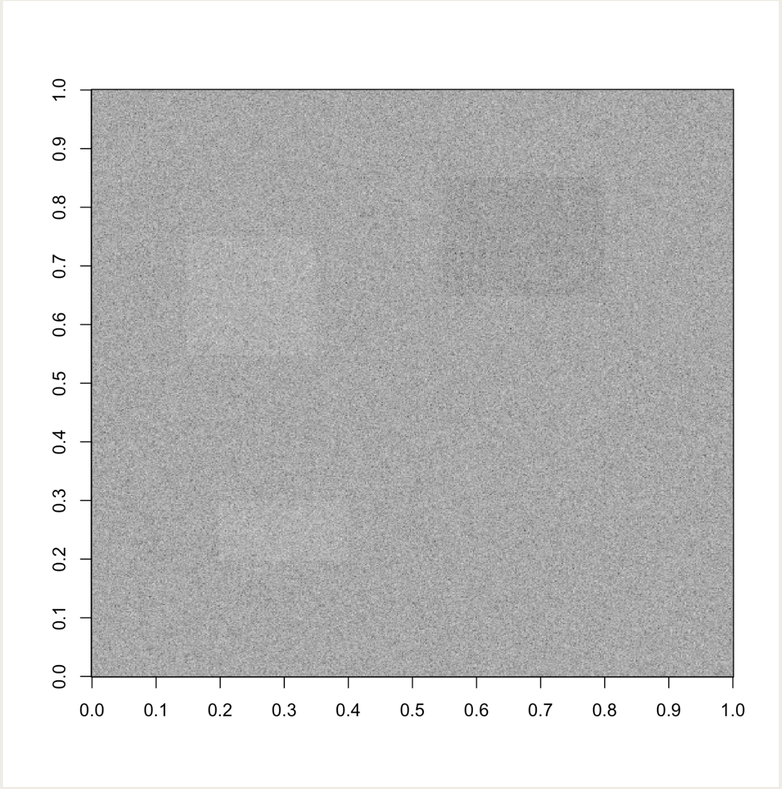
• To ensure that each connected components after first stage tests contain only one change-point patch , assume:

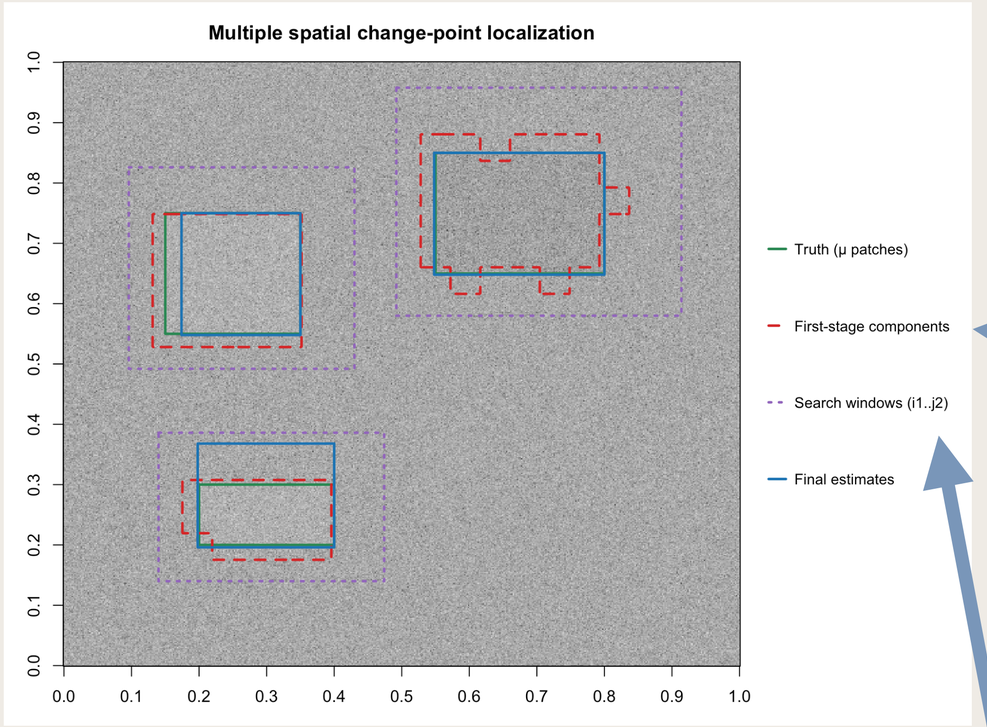
Spatial change-points are at least n_k^α-distance apart in at least one of the co-ordinates k. Here Γ_n has top right end-point $n = (n_1, \ldots, n_d)$.

Theorem *(Informal, BKM 2025+).*

If the j-th spatial change-point I_j has end-points a_j, b_j, then assume that $\min_{j \in [K]} |b_{jl} - a_{jl}| \gg n_l^\alpha \log^{1/d} n$ *for all $l \in [d]$. Then under above and additional technical but mild conditions,*

$$\mathbb{P}(\hat{K} = K, |I_j \Delta \hat{I}_j| > M_\varepsilon \delta_j^{-2} \text{ for } j \in [K]) < \varepsilon \,.$$



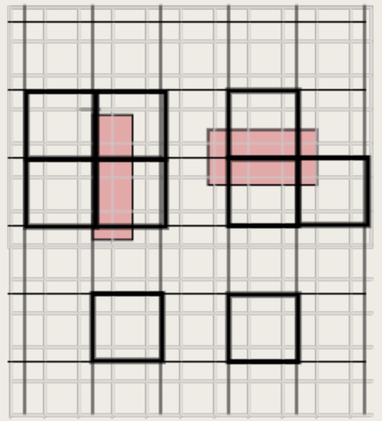


Γ_n : a 500×500 rectangle;

$I_1 = I_{[(0.2,0.2),(0.4,0.3]}$ with $\delta_1 = 0.5$.

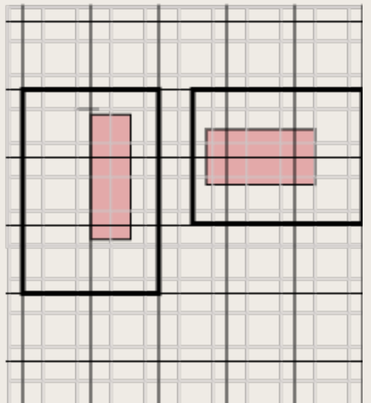
$I_2 = I_{[(0.15,0.55),(0.35,0.75]}$ with $\delta_2 = 0.5$.

$I_3 = I_{[(0.55,0.65),(0.8,0.85]}$ with $\delta_1 = -0.5$.



Blocks whose tests were rejected

Consider connected components of selected blocks. Remove smaller isolated components. Enlarge the remaining, irregularly – shaped components into rectangles.



Addendum

- Recall: The first stage tests require a *strong invariance principle*.

$$\max_{A \in \mathscr{A}} |S_n(A_n) - \sigma n^{d/2} \mathbb{W}(A)| = o_{\mathbb{P}}(n^{d/p}).$$

- Cut-offs require knowledge of σ.

How to estimate σ agnostic to presence of change-point patches?

For any X_i, the corresponding consistent sample mean estimate is:

$\bar{X}_{B_{k_i}}, k_i \leftarrow$ block containing i. Therefore, non-parametric consistent estimate is:

$$n^{-1} \sum_{s \neq t} (X_s - \bar{X}_{B_{k_s}})(X_t - \bar{X}_{B_{k_t}}) K\left(\frac{s-t}{B_n}\right),$$ where $B_n \to \infty$ with $B_n = o(n)$, K is some smooth kernel.


Questions?